AETHER-Q1

Quantum-Secured Hyperlink Communication Node

AETHER COMMUNICATIONS · CONCEPT DATASHEET

Communication beyond the horizon.

One node. Four spectra. Quantum-secured. Self-healing. Built for the places connectivity has never reached.

At a Glance

THROUGHPUT

1.2 Tbps terahertz fronthaul · **224 Gbps** LiFi · **multi-orbit LEO** uplink

SECURITY

Hardware **QKD co-processor** · NIST **post-quantum cryptography** · per-flow attestation

LATENCY

< 50 ms mesh self-heal · **< 8 ms** inter-satellite handover · sub-millisecond air interface

REACH

Operates from **-40 °C to +75 °C** · IP68 · ATEX Zone 2 option · MIL-STD-810H

INTELLIGENCE

256 TOPS edge AI · on-device 7B-parameter LLM · digital-twin synchronised

Why AETHER-Q1

The places that matter most — the deepest mines, the furthest platforms, the hardest disaster zones — are also the places radio struggles to reach. **AETHER-Q1** was engineered as an answer.

It is not another radio. It is the **convergence point** of four communication worlds, fused into a single weather-sealed node and orchestrated by an on-board AI that learns the spectrum around it. Where one band fails, another takes over. Where the network is attacked, quantum keys keep the channel honest.

> TIP
>
> **One node replaces a tower of compromise.** Operators who deploy AETHER-Q1 retire — on average — a base station, a satellite VSAT, a Wi-Fi backhaul, a SCADA radio, and a key-management appliance. *(Target reference architecture, Q1 2029.)*

Four Worlds, One Node

▸ Terahertz Fronthaul

The 0.30–0.45 THz band carries more data in a single beam than an entire 5G cell. AETHER-Q1 uses **holographic beamforming** assisted by intelligent reconfigurable surfaces, so the beam doesn't just point — it bends around obstacles. Target throughput: **1.2 Tbps**, line-of-sight, 100 m.

▸ Non-Terrestrial Uplink

A flat, electronically-steered phased array tracks **multi-orbit LEO constellations** across Ku, Ka, and V bands. No moving parts. No alignment drift. Inter-satellite handover **under 8 ms** — fast enough for closed-loop industrial control from anywhere on the planet.

▸ Optical Wireless (LiFi 2.0)

In hospitals, refineries, and secure facilities where RF is restricted, **light becomes the medium**. Visible-light and 850 nm IR backup deliver up to **224 Gbps** in a single emitter footprint — silent on the spectrum analyser, invisible to anyone outside the room.

▸ Self-Healing Mesh

Below the headline numbers, a **sub-6 GHz mesh** quietly knits together up to 4,096 nodes per cluster. AI-orchestrated routing reroutes around failure in **under 50 ms** — fast enough that a robot arm never notices the path it's using has just changed.

Quantum Security, Out of the Box

IMPORTANT

Tomorrow's adversary harvests today's traffic. AETHER-Q1 is built for the day quantum computers break the ciphers we trust now.

LAYER	MECHANISM	STANDARD
Key exchange	QKD co-processor (BB84 / E91)	ETSI TS 103 744
Symmetric crypto	ML-KEM-1024	NIST FIPS 203
Signatures	ML-DSA-87	NIST FIPS 204
Root of trust	Tamper-reactive HSM	FIPS 140-3 Level 4 *(target)*
Refresh rate	1 kHz key rotation	—
Architecture	Zero-trust, per-flow attestation	NIST SP 800-207

Performance

PARAMETER	VALUE	UNIT
THz aggregate downlink	1.2	Tbps
LiFi single-emitter	224	Gbps
Mesh nodes per cluster	4,096	—
Mesh convergence (failure)	< 50	ms
LEO inter-sat handover	< 8	ms
Edge AI compute	256	TOPS
Edge AI compute (FP16)	64	TFLOPS
Concurrent flows	1 M+	—

Form & Function

PARAMETER	VALUE
Dimensions	1,200 × 400 × 400 mm
Weight	28 kg (BASE) · 34 kg (EX)
Mounting	pole · mast · wall · vehicle roof
Power input	24–60 V DC / 100–240 V AC
Power draw	180 W typical · 420 W peak
Operating temp	-40 °C to +75 °C
Storage temp	-55 °C to +85 °C
Ingress protection	IP68 / NEMA 4X
Shock / vibration	MIL-STD-810H
Altitude	up to 5,000 m
EMI/EMC	MIL-STD-461G · EN 55032 Class A
Hazardous area	ATEX Zone 2 / Class I Div 2 *(EX variant)*

Interfaces

PORT	COUNT	NOTES
QSFP-DD (400 GbE optical)	2	Fronthaul / data plane
RJ45 10 GbE PoE++	4	90 W per port output
USB-C 4	1	Service & local console
M12 X-coded	1	Industrial Ethernet
CAN-FD	1	Vehicular variant only
SMA (GNSS / 1 PPS)	1	External time reference
Holographic ID	1	NFC + optical pairing

> **NOTE**
>
> A single hybrid fibre-and-power cable supports runs up to 1 km from the network edge — no separate copper, no roadside cabinets.

Built for the Frontier

Mega-construction. A Tbps backbone for autonomous earthmovers, drone surveys, and AR-guided crews — across sites measured in square kilometres.

Offshore energy. Quantum-secure SCADA back to shore, over LEO, with no exposed control plane on the public RF spectrum.

Mines & tunnels. Where radio dies, light keeps working. LiFi 2.0 carries traffic through galleries that have never seen a network bar.

Disaster response. Drop the node. Ninety seconds later, the mesh is alive. Rescue teams have a usable network before the dust settles.

Defence & contested theatres. Low-probability-of-intercept terahertz hops, anti-jam mesh routing, and one-time-pad-class key rotation — purpose-built for environments where the spectrum is the battlefield.

Smart ports. Deterministic networking from gantry to gate. AGVs, cranes, and customs systems on a single fabric, secured to a single root of trust.

Configurations

MODEL	BUILT FOR
AETHER-Q1-BASE	Standard industrial deployment
AETHER-Q1-EX	ATEX Zone 2 / IECEx hazardous area
AETHER-Q1-MIL	MIL-STD ruggedization, anti-jam firmware, classified-key option
AETHER-Q1-MED	Healthcare — EMI-quiet, LiFi-emphasised
AETHER-Q1-VEH	Vehicular roof-mount, CAN-FD bus, mobile mesh

Standards Targeted

3GPP Release 21+ · ITU-R IMT-2030 · ETSI TS 103 744 (QKD) · IEC 62443-4-2 · ISO 27001 · SOC 2 Type II · ANSI/ISA-95 · FCC Part 15 · ETSI EN 300 328

Roadmap

- **2027** — Engineering samples to lighthouse customers
- **2028** — Field trials: oil & gas · mining · smart ports
- **2029** — General availability · 1.0 THz extension module
- **2030** — Quantum-repeater integration for metro-scale entanglement

Reserve the Future

AETHER-Q1 will not be sold by the rack-unit. It will be sold by the mission. Tell us yours.

Engineering enquiries · engineering@aether.example **Reservation programme** · reserve@aether.example **Press & analysts** · press@aether.example

NOTE

Concept document. AETHER-Q1 is a forward-looking product concept aligned with publicly announced ITU-R IMT-2030 (6G) targets and current research in terahertz, LiFi, QKD, and post-quantum cryptography. Specifications are target performance for the indicated launch windows and may evolve. Items marked *(target)* are subject to certification at GA. Document AQ1-DS-v1.0 · 2026-05-09.